ZOO505

CELL AND MOLECULAR BIOLOGY

MIDTERM MEGA GRAND QUIZ FILE

LION ZOOLOGIST GROUP FOR 2nd SEMESTER

1. Water has tratomic molecule and consists of _ hydrogen and _ oxygen atom.

Ans. 2 and 1

2. _ can formed within RNA structure between the nearby self complementary sequence

Ans. Hairpin loops

3. For optimal activity lysosomes maintain a pH of about?

Ans. 5

4. Professional phagocytes ?

Ans. Neutrophils

5. Serve as the attachment surface for several structure including cell Wal glycolysis cytoskeleton ?

Ans. Plasma membrane

6. If stretch out some human chromosome would be several

Ans. Centimeters

7. Region of endoplasmic reticulum that lack _ are called smooth endoplasmic reticulum.

Ans. Ribosome

8. Pectin and hemicellulose of the cell wall in plants are synthesized in

Ans. Golgi Apparatus

9. In 1838 Mathias Shirleen a German botanist conclude that plants were made up of

Ans. Cells

10. Depending upon species and cell type the carbohydrates of the plasma membrane range between_ and _ percentage by weight.

Ans. 2-10%

11. DNA replicating during _ phase.

Ans. Synthesis

12. Power house of cell convert energy into useable from is called

Ans. Mitochondria

13. _ receive material from rough endoplasmic reticulum and modify them

Ans. Golgi Apparatus

14. The bacterial chromosome is as long as up to _

Ans. 1mm

15. Protein imported into the matrix of mitochondria are usually taken from _ within seconds or minutes of their release from ribosome

Ans. Cytosol

16. _ and the colleagues provide clue of structure of DNA

Ans. Erwin chargraf

17. Each ribonucleotide is composed of three component a nitrogen base phosphoric acid and _.

Ans. Ribose sugar

18. Glycogen hydrolysis secreting cells have abundant amount of_

Ans. SER

19. Phagosome usually has size of _ nm.

Ans. > 250

20. tRNA pair with the _ in such a way that amino acids are joined to growing polypeptide in the correct sequence.

Ans. mRNA

21. Statement incorrect about mitochondria?

Ans. It is single membran- enclosed organelle

22. Golgi Apparatus also lies on the exit route from _.

Ans. Endoplasmic reticulum

23. Three different RNA synthesis enzyme are known as

Ans. RNA polymerase

24. Plasma membrane play a critical role in the response of a cell to external stimulus a process known as

Ans. Single transduction

25. Carbon is _ element.

Ans. Tetravalent

26. In addition cell membrane are involved in a variety of cellular processes _ ?

Ans. Cell adhesion

27. Lysosomes are membrane enclosed compartment filled with _ enzyme that are used for the controlled intracellular digestion of macromolecules

Ans. Hydrolytic

28. During which relation water molecule is released?

Ans. Condensation

29. The electro static attraction is an important force in chromatin.

Ans. Stabilising

30. Water molecule move **from membrane by osmosis by pathways called**

Ans. Aqua porins

31. A phospholipids molecule travel from one into the other in about

Ans. One second

32. Study confirm that continued treatment of bacterial DNA with _ increase number of relaxed loops.

Ans. Nuclease

33. An egg has _ mitochondria .

Ans. A few hundred thousand

34. In the chromatin some of the loops super coiled and some are relaxed.

Ans. Bacterial

35. Lysosomal membrane protein are sythesis in rough endoplasmic reticulum and transport through _

In the lysosomes.

Ans. Golgi Apparatus

36. The process of forming mRNA on DNA template is called

Ans. Transcription

37. rRNA is a predominant material the ribosome consisting about _ of its weight.

Ans. 60%

38. There are _ type of RNA in living organism cells.

Ans. 3

39. The DNA connecting two nucleotide is called _ DNA

Ans. Linkers

40. The _ backbone of deoxyribose and phosphate group are out side the double helix facing the surrounding water

Ans. Hydrophilic

41. Bond between two hydrogen is

Ans. Covalent bond

42. The water generated in peroxisomes is rapidly broken down by the enzyme _ which is present in high concentration in these organelles

Ans. Catalase

43. In percentage world is in the midst of two scientist revolution one is information technology and other is

Ans. Molecular biology

4**4. Fusion of _ with phagosome required for autophagy**

Ans lysosomes

45. Control center of cell is

Ans. Nucleus

46. Mitochondrial precursor protein have a single sequence at their N terminus that is rapidly remove after importe by a protease

Ans. Matrix

47. Cholesterol absent from plasma membrane

Ans. Plants

48. RNA can be base pair with complimentary region of

Ans. Either RNA or either DNA

49. A reaction in which monmers one by one addition of water is

Ans. Hydrolysis

50. All histone have high percentage of

Ans. Arginine and lysine

51. The DNA isolate from different tissue of the same species have _ base composition.

Ans. Same

52. The single recognition particles that involved in the endoplasmic reticulum transport bind to the endoplasmic reticulum signals sequence as soon as a peptide bond merge from

Ans. Ribosome

53. Sepecialii organelle for aerobics respiration in animals

Ans. Mitochondria

54. Bacterial chomatin contains multiple loops which emerge from

Ans. Central

55. In the DNA structure the two antiparallel strands for double helical DNA are _ to each other

Ans. Complimentary

56. Most membrane are phospholipid due to

Ans. Phosphate group

57. Plants seedlings relay on the stored _ to provide the energy to material from new plant

Ans. Faty acids

58. Signal recognition particles consists of six different _ chains.

Ans. Polypeptide

59. Which group variable in amino acids

Ans. R

6**0. The term molecular biology substituted for another term**

Ans. Molecular genetic

61. Each neucleousom contain approximately _ no of DNA

Ans. 200

62. Cell division distinct from nuclear division and occurs at the end of _ phase.

Ans. Interphase

63. _ RNA component of ribosome

Ans. Ribosomal RNA

64. _ German zoologist published a comprehensive report on the cellular bases of animals life

Ans. Theordor Shawn

65. tRNa serves as _ molecule in the protein synthesis

Ans. Adapter

66. During endoplasmic reticulum the translocator complex form a water fiilled pore in the membrane through which polypeptide chains transverse the it consists of 4 _.

Ans. Protein

67. _ transport two different substance in the opposite direction

Ans. Antiport

6**8. In the Golgi Apparatus the ERGIC53 serve as for the packing some secretary protein into COPII coated vesicles**

Ans. Receptors

69. Gene are physical unit located on

Ans. Chromosome

70. Cell that do not divide are usually arrest in _ phase

Ans. Interphase

71. General formula of carbohydrates

Ans. CH2O

72. Chloroplasts contains their own DNA and ribosome for making some ot their own

Ans. Protein

73. During endoplasmic reticulum transport an endoplasmic reticulum signals sequence is recognize

Ans. Twice

74. Most abundant molecule in cell

Ans. Protein

75. Lysosomal hydrolases Cary a unique marker in the form _ group

Ans. Manose 6- phosphate

76. The cell are separate from external world by thin fragile structure

Ans. Plasma membrane

77. SPR consists of six different chains to bound to a single small _ molecule

Ans. RNA

78. **Garbage disposal** done in

Ans. Lysosomes

79. Nuclear division occurs during

Ans. Mitosis

80. The nitrogen base of the strand of DNA structure are stretch inside the double helix lying _ to the long axis of the helix.

Ans perpendicular

81. _ transport **substance** in one direction

Ans. Uniport

82. Peroxisomes are simple membrane bound vesicles with diameter of _ micrometer

Ans. 0.1 to 1.0

83. The more extended form of chromatin is called

Ans. Euchromatin

84. Structure that degrade unwanted material in the cell

Ans. Lysosomes

85. Plasma membrane is only _ wide.

Ans. 50 to 10 NM

86. A less abundant class of membrane lipids called

Ans. Sphingolipids

87. One of the primary metabolic activity of _ germinating seedlings in the conversion of stored faty acids to carbohydrates

Ans. Glyoxisomes

88. Sphingolipids derivative of

Ans. Spigosine

89. Organism shuffle DNA mostly during

Ans. Meiosis

90. Histone protein have _ major class

Ans. 5

91. In Golgi Apparatus the ERGIC53 serve as a _ form packing some secretary COPII coated vesicles

Ans. Receptors

92. Phagocytosis involved the ingestion of large particles microorganisms or dead cell via large vesicles called

Ans. Phagosome

93. Cytoskeleton consists of

Ans. Microfilaments

94. The neucleousom particles consists of an _ protein complex.

Ans. Octameric

95. If super could DNA molecule of a bacteria receive a Nick the strains under winding is immediately removed and super coiled is

Ans. Complete list

96. Each neucleousom contain _ histone

Ans. 9

97. Fusion of _ required for autophagy.

Ans. Lysosomes

98. Electrostatic attraction is an important _ force in chromatin

Ans. Stabilising

99. A typical human cell contain about _ million copy of each kind of histone

Ans. 60

100. Thymine is replace by _ in RNAs.

Ans. Uracil

101. Pinocyt involve the ingestion of

Ans. Fluid and solutes

102. There are more than _ mitochondria in liver.

Ans. 1000

103. The ERGIC53 is a _ that bind manomers on factor v and factor vii protein

Ans. Lectins

104. The bond forming by sharing of electron is called

Ans. Covalent bond

105. Primary active transport required energy

Ans. ATP require directly

106. Misfolded assemble are retain in the endoplasmic reticulum where they are bound to _ protein.

Ans. Chaperone

107. The plant cell Wal has

Ans. Cellulose

108. Peroxisomes have multi function organelle contain more than _ enzyme.

Ans. 50

109. Head of membrane phospholipid

Ans. Polar hydrophilic

110. During which reaction water released

Ans. Condensation

111. All the professional phagocytes develop from

Ans. Homeopathic stem cell

112. To exist from the endoplasmic reticulum protein must be properly folded and completely

Ans. Assembled

113. Fluid mosaic model proposed in 1972 by

Ans. S. Janther Singer and Garth

114. The reverse of consent reaction

Ans. Hydrolysis

115. In the DNA the number of adenine is equal to the number of

Ans. Thymine

116. More than _ percentage of the membrane carbohydrates is c bond to protein to form glycoproteins

Ans. 90%

117. How many types of lipids present

Ans. 3

118. The DNA connecting two neucleousom called

Ans. Linkers

119. Peroxisomes may contain a dense crystalline core of enzyme

Ans. Oxidative

120. Proteases are

Ans. Protein digesting enzyme

121. One of the following did not help to provide a evidence that DNA rather than is herdatery material

Ans. Lois pasture experiment

122. _ transport two different substance in Same direction

Ans. symport

123. The chromosome of eukaryotes are large and more _ than those of prokaryotes

Ans. Complex

124. In _ complementary human genome has been sequence

Ans. 2003

125. Macrophages phagocytes more than hundred billion _ cell every day

Ans. RBCs

126. Endoplasmic reticulum in most eukaryotic cell store

Ans. Ca2+

127. Enzyme of peroxisomes are involved in oxidation _ faty acids

Ans. Very long chains

128. Tails of membrane phospholipid are

Ans. Non.polar hydrophobic

129. The tendency of atom to atom to attract electron towards itself is called

Ans. Electronegativity

130. _ transport two different substance opposite direction

Ans. Antiport

131. In 1935 _ proposed the primary membrane composed of a lipid bilayer

Ans. Hugh Davison and James

132. The Golgi complex major site of _ synthesis

Ans. Carbohydrates

133. Pinocyt make vesicles of _ diameter.

Ans. 100nm

134. _ manufacture polysacride for the plant Wal

Ans. Golgi Apparatus

135. _ of cell die each day in human for example epithelial cell

Ans. Billion

136. Golgi Apparatus also lies on the from the

Ans. Endoplasmic reticulum

137. _ postulate the three dimensional structure of DNA

Ans. James Watson and France Crick

138. Cell with two copies of gene is called :

ANS : Diploid

139: Endoplasmic reticulum , an organelie made up of ,

ANS : Two sub-units

140 : The reverse of condensation reaction is ……

ANS : Hydrolysis

141 : ……Cells have cellulose containing cell walls

ANS : Eukaryotic

142 : Peptidase enzymes in involved in the breakdown of :

ANS : Proteins

143 : Partly smooth and partly rough region of ER are called :

ANS : SER

144 : Mitochondrial proteins are first fully synthesized as precursor protein in the …….. & then translocated into mitochondria .

ANS : Cytosol

145 : The …… backbone of alternating deoxyribose and phosphate groups are on the outside of the double helix of DNA :

ANS : Hydrophilic

146 : A function of microtubules includes :

a. Support the cell
b. Act as tracks in the cell
c. Resist compression
d. All of these

147. Which of the following is a type of endocytosis ?

ANS : Pinocytosis

148 : The bacterial chromosomes is looped and folded and attached at one or several points to the ……. Membrane .

ANS : Plasma

149 : Which of the following are eularyotes

a. Plants
b. Animals
c. All
d. Fungi

150. The signal-recognition particle (SRP) consisting of six different polypeptide chains bound to a single.

ANS : small RNA molecule

151 : If a super cooled DNA of bacteria receives a nick , the stain of under winding is immediately removed and the super coiling is …

ANS : Completely lost

152 : RNA (ribonucleic acid) is a polymer of .

ANS : Ribonucleotides

153 : Mitochondrial proteins are first fully synthesized as precursor proteins in the …….. & then translocated into mitochondrial

ANS : cytosol

154 : Ribosomes are either free or bound to

ANS : Endoplasmic reticulum

155 : A posttrsnslational process involves the transport of protein into …….

a. All of the bellow
b. Nuclei
c. Mitrochondria
d. Chloroplasts

156 : Bacteria typically have ……. Chromosome cinsisting of DNA with associated protein .

ANS : single circular

157 : The positively charged side of chains of lysine and arginine histone to bind to the negatively charged …… groups (..)

ANS : Phophate

158 : The term related to genes product and heredity is called …..

ANS : Molecular biology

159 : Which gives mechanical support to the cell and help to maintain its shapes ?

ANS : cytosleleton

160 : One of the following did not heko to provide evidence that DNA rather than protein is the hereditary matter

ANS : Louis Pasteur experiments

161 : Co-translational process of protein occurs in the ….

ANS: Endoplasmic reticulum

162 : serve the attachment surface for several structures , including the cell wall glycocalix and cytoskeleton

ANS : plasma membrane

163 : perocxisomes got their name because they are the site of synthesis and degradation …… a highly react..)

ANS : Hydrogen peroxide

164 : Bacterial chromatin consist of multiple loops which emerge from a ……. Region of the chrom…

ANS : central

165 : Microfilament are composed of .

ANS : Actin protein

166 : Cyanobacteria chlorophyll containing have two folds of plasma membrane other have …..

ANs : Mesosomes

167 : A nucleoside is a part of

ANS : Nucleotide

168 : A reaction in which monomers released one by ine by addition of water is termed as ……

ANS : Hydrolysis

169 : 18-misfolded incompletely assembled are retained in the ER , where they are bound to ……. Proteins

ANS : chaperone

170 : Cyanobacteria chlorophyll containing have folds of plasma membrane , other have ……

ANS : mesosomes

171 : Erwin chargaff and his colleagues provided a most important clue to the structure of DNA in …

ANS : 1940s

172 : Chemically modification of proteins (glycosylation) take place by .

ANS : SER

173 : Each nucleosomes contains …. Histone

ANS : 9

174 : In present age world is in the midst of two scientific revolutions , one is information technology and other is …..

ANS : molecular biology

175 : James Watson and francis crick postulated a ….. dimentional model of DNA structure in 1953

ANS : Three

176 : There are mainly ……. Types of ribonucleic acids (RNA) present in the cells of living organism

ANS : 3

177 : Peroxisomes are multifunctional organelles contgaining more than ….. enzymes

ANS : 50

178 : Each nucleosome is formed ny winding DNA fibre around a protein assembly consisting of ….. histone molecule .

ANS: 8

179 : The different substances transported in same direction across cell membrane is :

ANS : Symport

180 : All proteins transported via endoplasmic reticulum have ….. signal sequences

ANS : Multi

181 : Smooth endoplasmic reticulum takes …. On its surface .

ANS: Ribosomes

182 : Proteasomes are ….

ANS : protein digesting enzymes

183 : The Wall of the hollow tube microtubule is composed of globular called :

ANS : Tublin

184 : ……manufacture poly-sacchrides for the plant wall .

ANS : Golgi apparatus

184 : ……contain 40 types of enzymes , including protease , nucleases , phosphatases and sulfatases they are like are acids hydrolase

ANS :; Lysosomes

185 : Some protists have ……..number of mitochondria .

ANS : 1

186: What are the functions of cytoskeleton ?

a. Keep in place the organellele
b. To change the shape of cell
c. ANchar cytoplasmic enzymes
d. All the above

187 : The cell are separated from the external world by a thin , fragile structure called …..

ANS : plasma membrane

188 : Water is triatomoc molecule and consist of …hydrogen atom and … oxygen atom .

ANS: 2,1

189: Partly smooth and partly rough region of ER are called

ANS ":SER

190 : During prophase -1 of meiosis the homologous chromosomes line up to form

ANS : Tetrad

191 : The tubules and sacs are interconnected the ER membrane enclose..

ANS : ER cisternal space

192 : plasmid is the extra chromosomal DNA in

ANS : Plants

193 : Engulfing of bacteria by white blood cells is called as …..

ANs : phgocytosis

194 : The ER membrane sperates theER lumen from the …..

AnS : cytosol

195 : Smooth endoplasmic reticulum lacks ……..on its surface

***ANS:* Ribosomes**

196 : Ribosomes are assembled in the …

ANS : Nucleolus

197 : The nitrogenous base of both starnds of DNA structure are staked inside the double helix lying ….. to the long axis of the helix .

ANS : perpendicular

198 : During ER transport the translocator sec61 complex forms a water filled pore in the membrane consist of a ….

ANS : Proteins

199 : Purine is double ring structure binds to the …… of pentose sugar

ANS : Carbon#5

200: In which process the fibres of the cytoskeleton are involved ?

ANS : both b and c

201 : Input of energy is not required in

ANS': Passive transport

202: There are many three types of ribonucleic acide (RNAs)

ANS ;; mRNA ,rRNA , tRNA

203 : …… RNAs are components of ribosomes

ANS : rRNA

204 : In the DNA strycture the two strands of DNA are present in ……. Orientation i.e, their 5,3-phosphodiester bonds run in the opposite direction

ANS : Antiparallel

205 : ATP is utilized directly in

ANS : tertiary active tansport

206 : sphinogolipids are derivatives of ……

ANS : spingosine

207 : Digestion of foreign particle by enzymes occurs in the

ANS : lysosomes

208 : In DNA the number of adenosine residues is equal to the number of ……..

ANS : Thymidine

209 ; Accroding to chargraff rules . the some of the purine residues …….. the sum of the …)

ANS : Equals

210 : Unicellular organism propagate by splitting into …. Cells .

ANS ; 2

211 : Which of the following are not synthesized by the endoplasmic reticulum?

ANS : Cholestrol

212 : Anton van leeuwenhoek examine a drop of pond water under the microscope and he observed the crawling microscopic entitles which he ..)

ANS : Animalcules

213 : Genetic material must :

a. Encode phenotype
b. Replicate faithfully
c. All of these
d. Contain complex information

214: ……. Of cells die each day in humans for example blood epithelial cells .

ANS : Billions

215 : Highly condensed chromatin is known as :

ANS : Heterochromatin

216 : specific protein interact with the bacterial DNA to form a highly condensed ……

ANS : nucleoprotein

217 : Lysosomes are members-enclosed compartments filled with ……..enzymes that are…)

ANS : Hydrolytic

218 : Ribosomes are made up of proteins and ……

ANS : mRNA

219 : Signal sequences were first discovered in ……. Imported into the rough ER

ANS : Protein

220 : The first proposal that cellular membranes might contain a lipid bilayer was made in …… ny two dutch scien.....)

ANS : 1925

221 : Macrophage phagocytose more than hundred billion …… cells every day .

ANS : RBCs

222 : SRP – ribosome complex binds to the SRP receptor . this interaction brings the SRP-ribosome..)_

ANS : protein translocator

223 : Pectin involved in the cell wall of plants is manufactured in

ANS : Golgi apparatus

224 : Heads of ,membrane phospholipids are ,……..

ANS : polar hydrophilic

225 :A chromosomes is an organized structure of …. And …..

ANS : DNA , RNA

226 : Cilia and flagella help in the …. Of cell

ANS : movement

227 : What is the diameter of the intermediate filament

ANS : 10 -12 nm

END……